02. Payments Solution design

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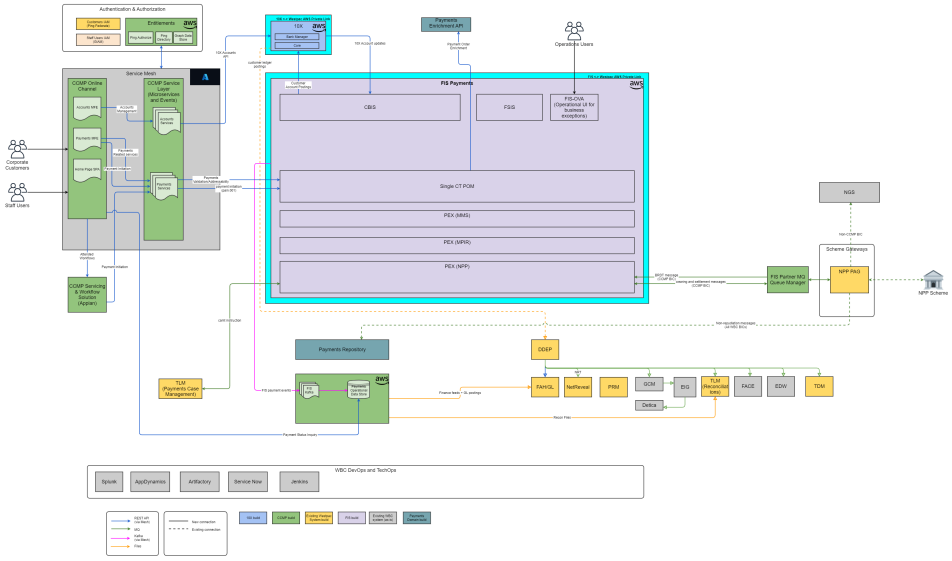
ACD References

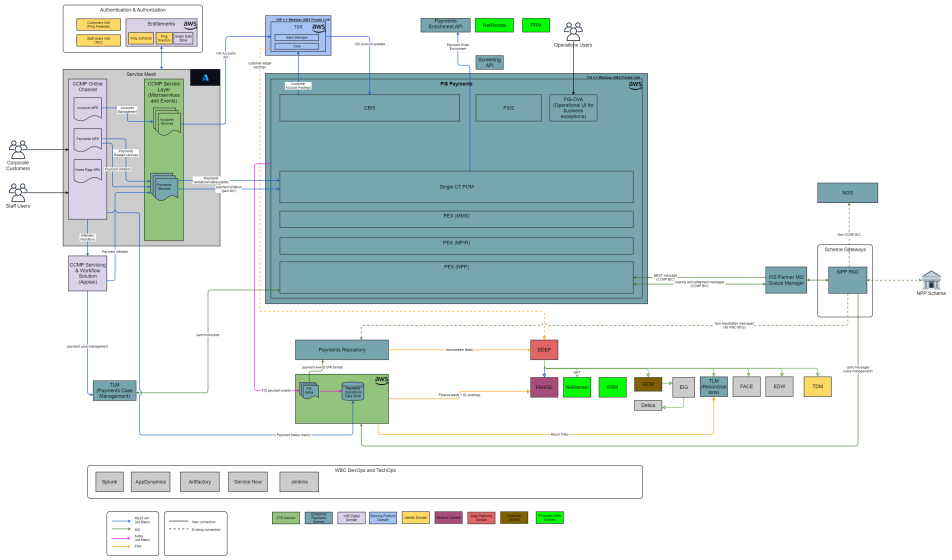
1. CCMP Payments - FIS Hosting ACD
2. 3. ACD for 10x Integration (Architecture On a Page)
3. CCMP Data - Data Solution ACD
4. W1C Payment Mesh Portfolio

Diagram

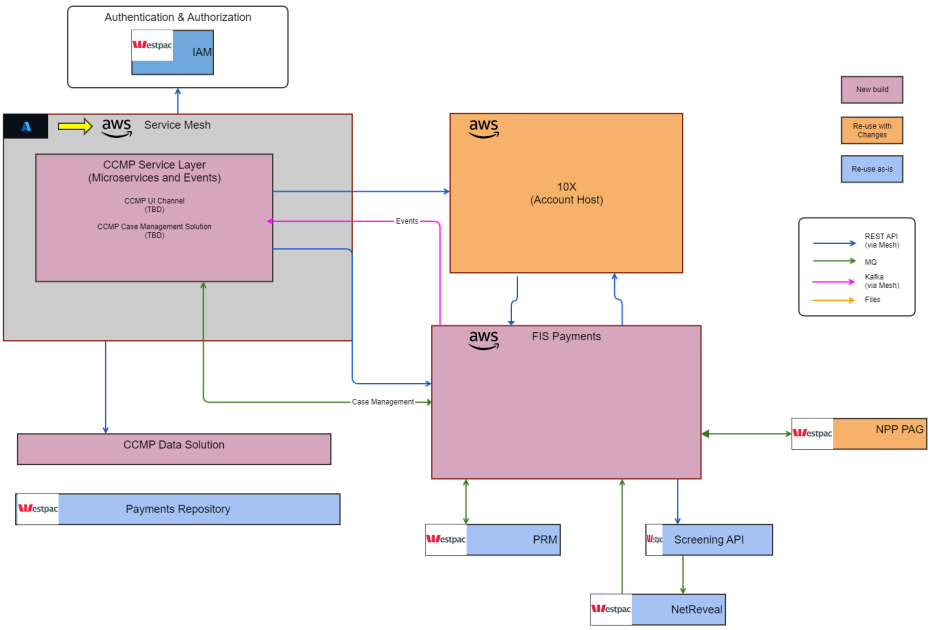
HL diagram for Beta State:



Domain View - WIP:

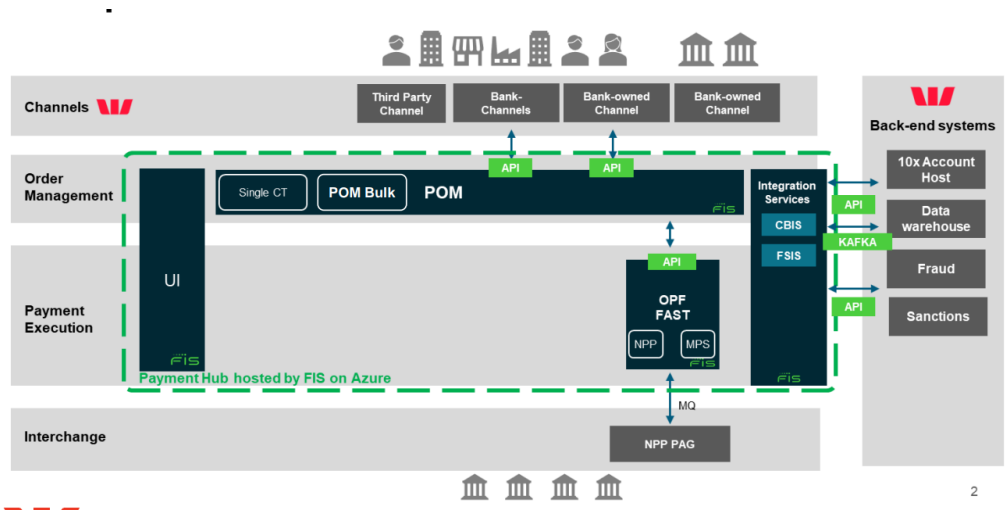


Simplified view



there is an indirect connection atheist from FIS into IAM for BVA. For API flows no connection to IAM needed

FIS



High-level description of the FIS components

FIS dictionary

Pre-requisites

Account cache a component within CBIS FIS (A013D9) called Account cache. The account cache can be updated by batch or api: Account Cache Update High Level

Account cache has the following usages:

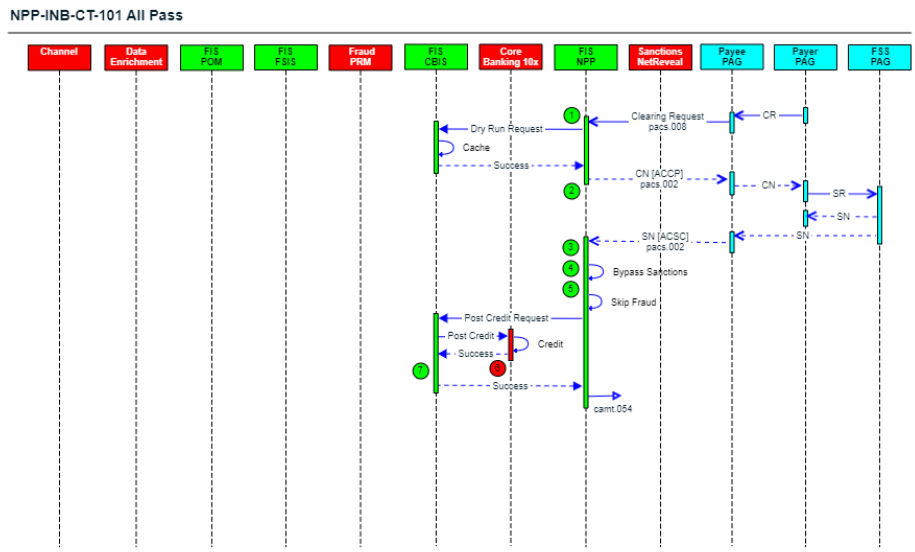
1. Account Location lookup service uses the account cache to find out which account host the account belongs to when they decide which adapter to use for the posting.
2. Eligibility check:
    a. Whether the account can receive account postings
    b. Whether the account is eligible for the payment scheme
        i. Extra check for NPP: Whether the account is eligible for the related Business service.
3. Account number and BSB to Subscription key lookup
    a. Subscription key is an internal identifier within 10X used to identity an account. Usages identified:
        i. Account Posting

- 01. SD: Account Cache - API for Create and Update
- 02. SD: Account Cache - Batch synch and Re-synch to fix data discrepancies
- Cache vs 10X directly
- Detailed NFR: for Account Cache

To-Us NPP

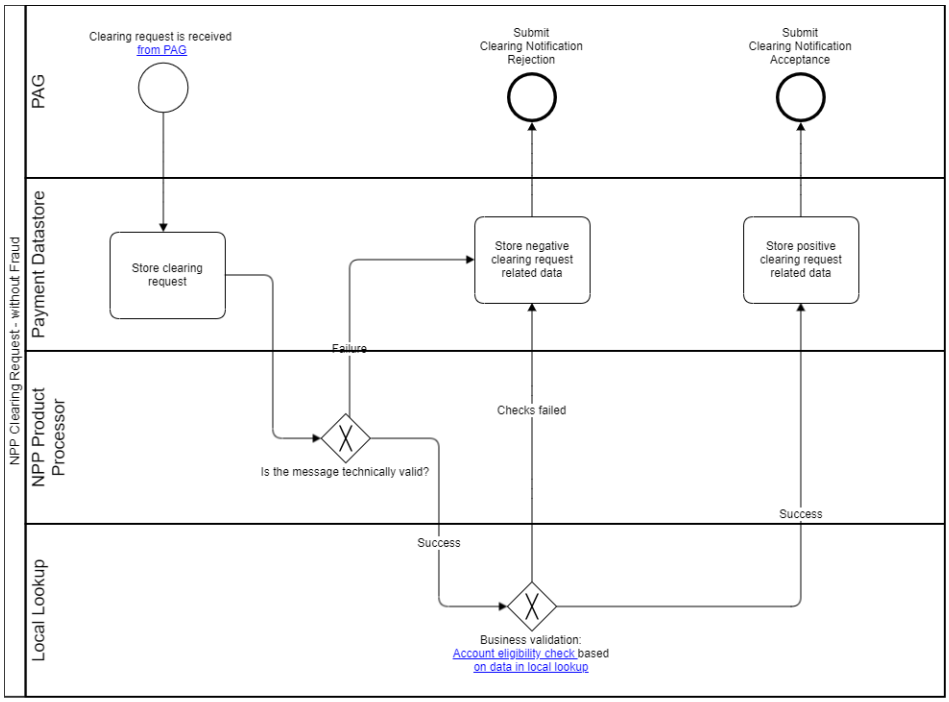
Happy Path Sequence Diagram

NPP-INB-CT-101 All Pass

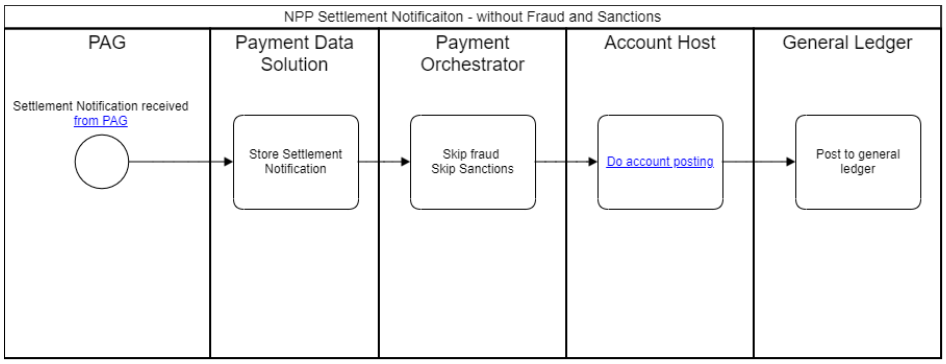


To see what the numbers mean go: NPP-INB-CT-101 All Pass

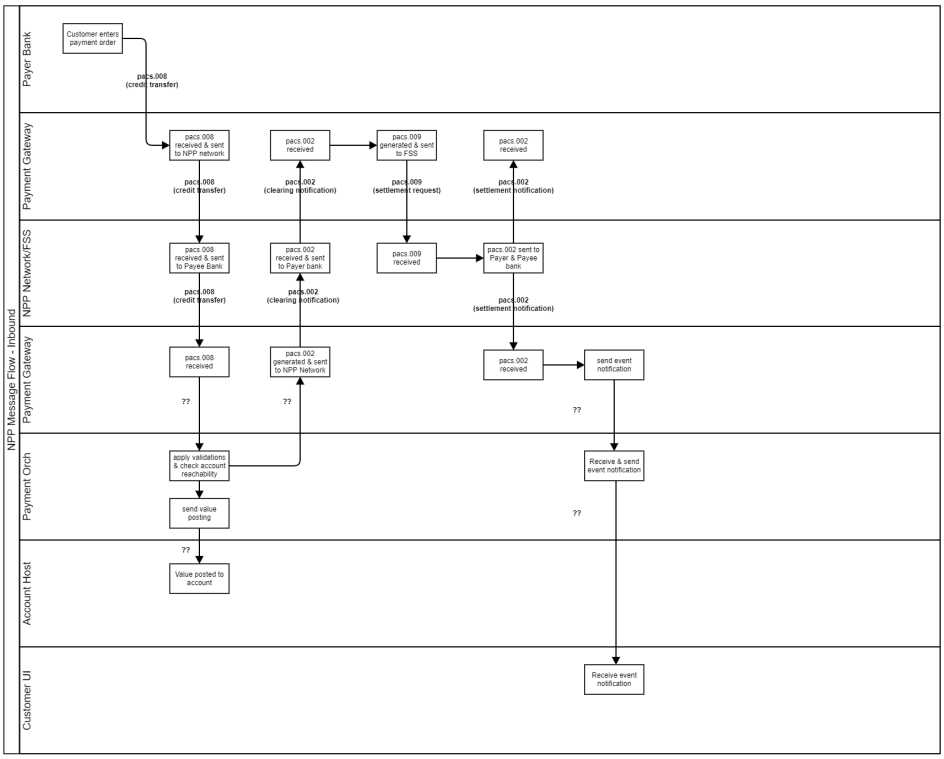
Clearing Request To-Us



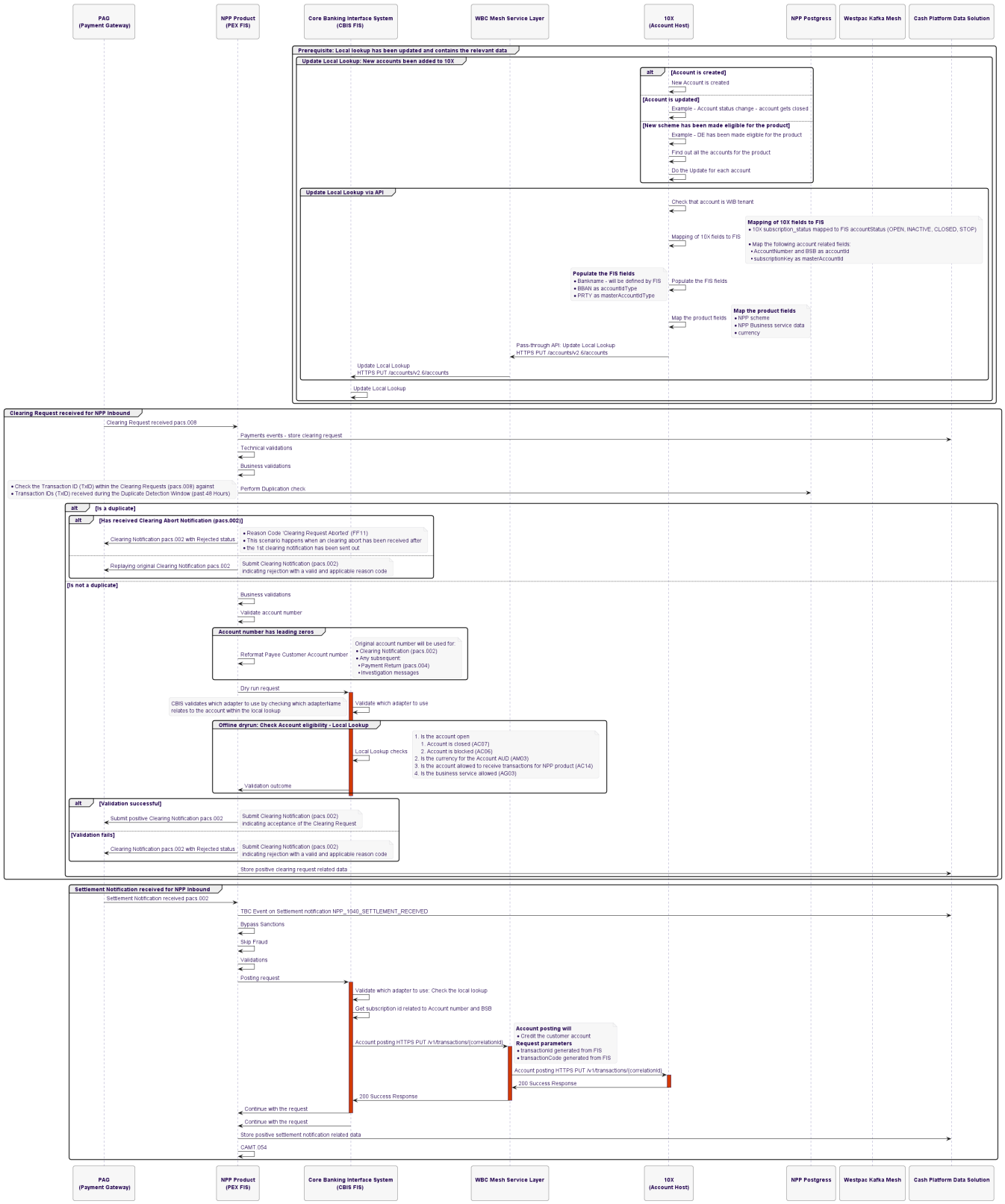
Settlement Notification To-Us



NPP Message Flow - To-Us



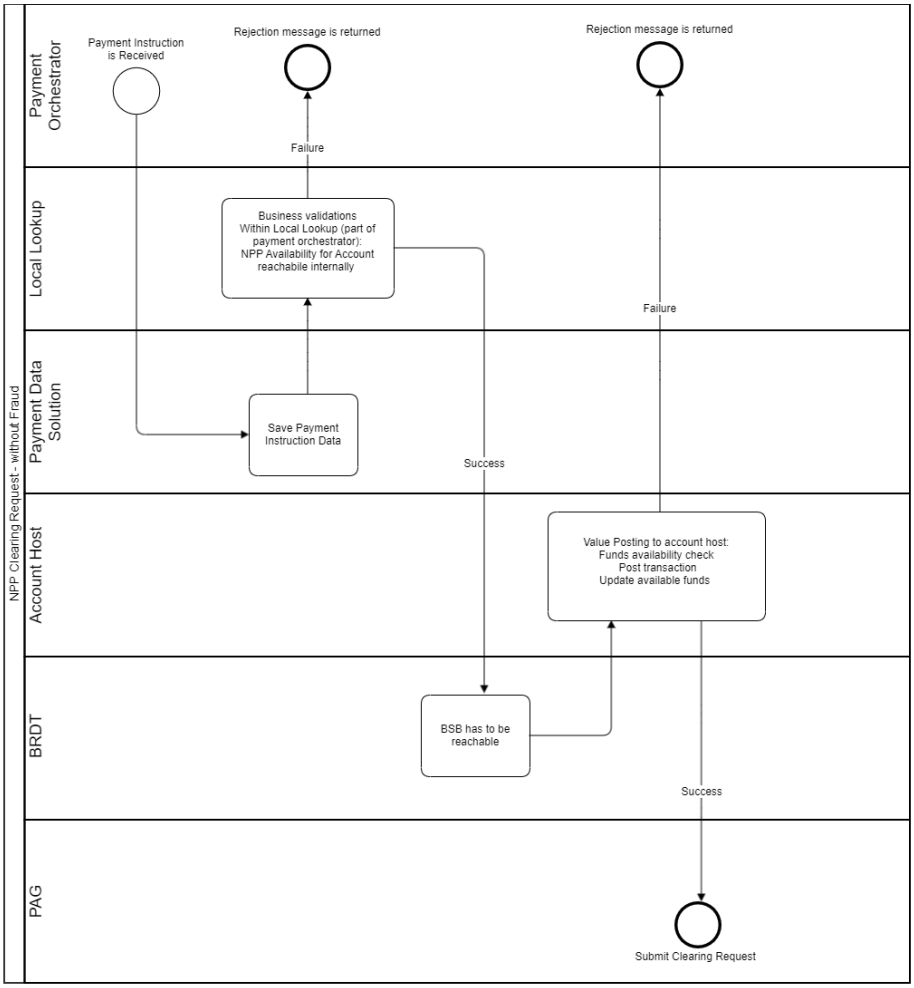
To-Us Sequence diagram



On-Me

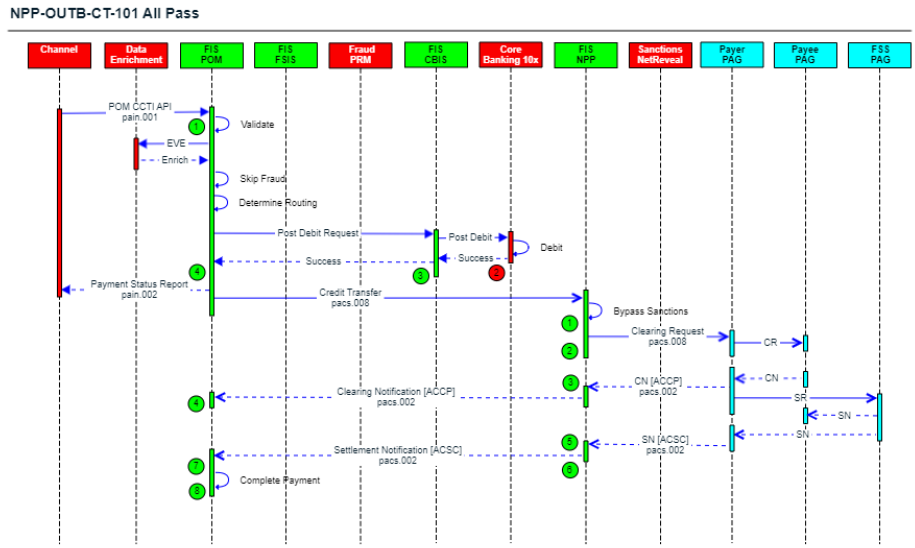
Unable to render {include} The included page could not be found.

Off-Us NPP



Sequence Diagram

NPP-OUTB-CT-101 All Pass



To see what the numbers mean go: NPP-OUTB-CT-101 All Pass

Systems

10X High-level

10X (A00A46) Account host and Payment Engine for DE. Is the customer ledger. Has multiple tenants, there is a specific tenancy setup for WIB - Cash Platform.

Account Cache High-level

Account cache a component within CBIS FIS (A013D9) called Account cache. The account cache can be updated by batch or api: Account Cache Update High Level

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ATLAS - High-level

ATLAS (A016F9) (Accounts & Transactions & Ledger Aggregated Store) is an Operational Data Store modelling data from 10x. It curates product, subscription and transactional data for consumption by channels and for downstream integration for statements and other processes.

BIS High-level

BIS BIS gets files from applications and pass them on to to Trading Bank, so is just an interface.

BVA (NPP) High-level

BVA (NPP Product) Bank Visibility Application - the UI for NPP Product from FIS (A013D9).

CBIS High-level

CBIS Core Banking Integration Services - a product solution from FIS (A013D9). For more detailed information see: Core Banking Integration Services (CBIS - FIS Product Solution)

- Manages integration of the payments systems to talk to one or more core banking systems - can talk to 10X.
- Provides functionality for account lookup/cache/ stand-in for the core banking systems.
- Integrates with any payment system (i.e. NPP / POM) and any core banking solution.
- Acting as an integration layer, the CBIS provides banks a cost-effective way of decoupling management and execution of payments from actual posting to accounts.
- Product (internal reference to CBIS) Adaptors by FIS - standard product interfaces for performing some specific business functions. , for more information see: CBIS Adaptors. Some examples below:
    - PRA - Product Rules Adapter provides a mechanism for CBIS to better determine posting eligibility for posting to an account.

    - PBA - Product Behaviour Adapter provides a mechanism for CBIS to better determine the correct behaviour for several functionalities for one individual accounting entry.
    - AVA - Account Validator Adapter provides a mechanism for CBIS to execute validations against provided accounts via Account Eligibility Entries.
- Custom Adapters by FIS - can be used to customise interfaces when transformation is needed. An example is a 10x adapter.
- CBIS CORE - this cannot be changed. It's all code and no configuration.

DDEP High-level

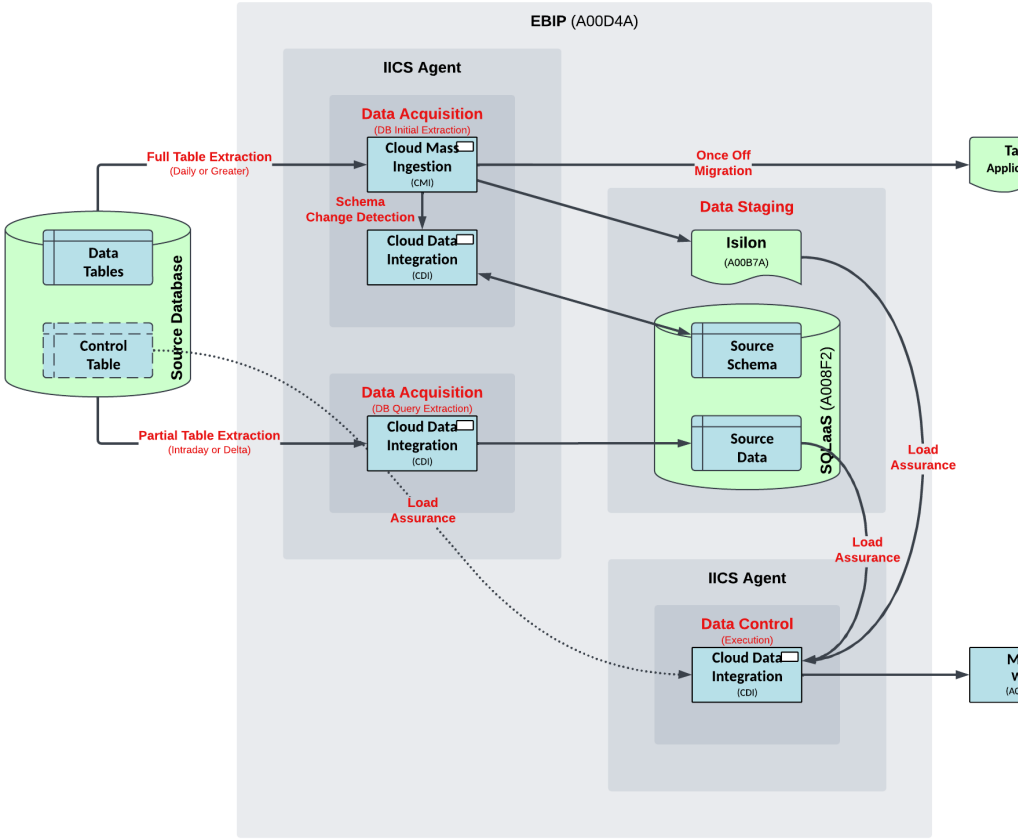
DDEP (A006B2) Data warehouse and file processing. DDEP Framework

DE High-level

Direct Entry (A00003) Transfers DE files to the coin network. Routes W1C BSBs to 10X for DE payments to be transferred as an inward type 1.

EBIP - High-level

EBIP Enterprise Batch Integration Platform application is also used to perform data extraction operations. It is powered by Informatica IICS using both Cloud Data Integration (CDI) and Cloud Mass Ingestion (CMI) services. CMI service is designed for modern ELT processes to cloud data warehouse or lake-house. See here for more information -- Batch Integration - Database Extraction Solution Template



EDW High-level

EDW (A0036B) have three layers:

1. Source image (exact source copy)
2. Core (consumption layer, foundational data model with all the data available, more basic structure)
3. Semantic model (very specific data model layer, this is where other systems interfaces and comes to consume the data)

EIG High-level

EIG (A00278) Does File transfer and transformations. Is used heavily in the TLM flows.

FAH High-level

FAH (A00506) Oracle Financials Account Hub. Transactions and Balances gets passed to FAH. FAH receives every single transaction (FAH maintains all the transactions at an individual level), the only exclusion is accrued interest from 10X since they were causing rounding issues. These individual transactions are mapped to a line item in the GL and is posted as a consolidated posting (FAH contains aggregating logic to help with this and mapping rules to derive the GL code block). FAH requires a balance file and that is why the balances are also passed to FAH.

FTH High-level

FTH Account host isolation and payment-pre-processor. Does Fraud, Credit check, balance check, account validation as a part of their FTH Generic Payment Flow. For account host isolation it creates BIS files for Trading bank posting, does Memo posting to AO (later synched overnight to Trading bank), Posting for AGSL accounts.

GL High-level

GL General ledger. GL maintains the balance and gets passed transactions.

MMS MPS Product High-level

MPS Product (MMS) Mandate Payment Services (Mandate Management Services) - is a component within the FIS MPS product that supports the Mandate management functionalities. MMS is the Mandate Management Service that is a centralised service provided by the NPP Basic infrastructure. This components supports functionalities for managing mandates, such as:

- Create
- Amend
- Recall
- View

PAG High-level

PAG (A00605) Payment Gateway (PAG) is the interface between a Participant's back-office application and the Basic Infrastructure. Payment Gateway exchanges ISO 20022 messages with a Participant's back-office application over IBM MQ. Payment Gateway also provides a SOAP-based interface between the Participant's back office application and the Addressing Service for query and update processing. Payment Gateway manages the orchestration of the clearing and settlement flow between Participants and the Reserve Bank of Australia (RBA). Payment Gateway manages processing of other messages exchanged using the basic messaging flow.

PEX(NPP Product) High-Level

PEX (NPP Product) Payment Execution System for New Payment Platform - scheme specific back-office execution systems. For NPP there's an NPP specific Payment Execution System that is also referred to as NPP Product by FIS (A013D9).

PODS High-level

PODS (A0154A) Payment Operational Data Store - FIS Payments stores payment history for only 28 days. The purpose for PODS is to ingest payment events from FIS to keep a long history and a fully reconciled and persistent state for payments.

Services for data include:

- Payment status and history from FIS
- Limited information from 10x (Enrich payment information)
- Event matching and data reconciliation
- End of Day posting to FAH and GL for NPP
- Support audit, investigations and ad hoc customer enquiries

POM High-level

POM Payment Order Management - a product solution from FIS (A013D9) with capabilities to manage payment initiation and related business processes. Provides capabilities to process payments and related messages, originated from any channel and destined to any clearing and settlement execution system. Scheme agnostic.

- Receives payment requests from front-office or back-office channels via REST APIs
- Prepares payment for execution
- Orchestrates posting of funds, when required
- Selects the appropriate payment execution system (PEX)
- Routes and submits payment to the payment execution system (PEX)
- Provides Payment Status Reports to the originator

Qvalent high-level

Qvalent Third party partner fully owned by GTS in WBC. Has the following key components:

- Gopher is for integration with Mesh API services
- Quickstream is UI for institutional customers
- Payway is UI for business banking customers